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Final Term Assignment

1. Information Retrieval

Information Retrieval (IR) is the process of finding relevant data from a collection of information from large databases. This is used in search engines like Google, where you type in a question or keyword and it returns results that are most likely to be useful. For example, when you search for best laptop under 50,000 Tk the search engine uses information retrieval techniques to find the most relevant pages.

2. Cloud Database

A cloud database is a database that is hosted and accessed over the internet ~~int~~ instead of being stored on a personal computer or physical

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server. This means the data is stored in the cloud and can be accessed anytime, anywhere, as long as you have internet access. Example like Google Cloud SQL, Amazon RDS, or Microsoft Azure SQL Database. It is helpful because it makes data storage and management easier and more scalable without needing a physical server.

3. Graph Database

A graph database is a type of database that uses graph structures to store data. This means it organizes data into nodes (points) and edges (connections). For example, in a social media network, the nodes could represent people and the edges could represent their friendships.

Graph databases are useful for handling data

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that involves connections like social network or Netflix, YouTube etc.

4. Distributed Database

A Distributed Database is a database that is spread across multiple locations or servers. Instead of having all the data stored in one place, it is distributed across different machines. It might be in different parts of the world. This helps in improving the performance, reliability and availability of the data. For example, when you use an online service like Google Drive. The data is stored in different data centers to ensure it is always available even if one server goes down.

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5. Bioinformatics

Bioinformatics is a field that combines biology, computer science and information technology to analyze and interpret biological data. It is used to understand genetic sequences, protein structures and other biological information. For example, scientists use bioinformatics to understand how genes affect ~~de~~ diseases or to help design new medicines. It ~~invotes~~ involves using algorithms and software to process complex biological data.